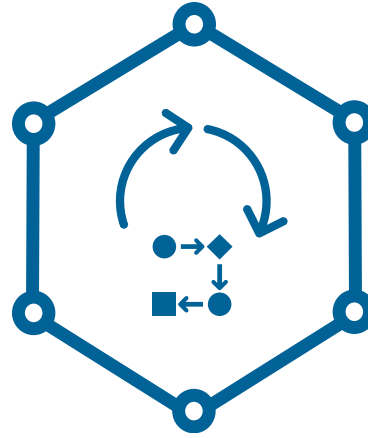




Knowledge Graph Evolution Schema Evolution

Bonus

Emanuel Sallinger



STANFORD

Slides courtesy of Vinay K. Chaudhri
University of Stanford

Schema Evolution

- For a relational database
 - Adding/removing a column, renaming an attribute
 - Known as database reorganization

Schema Evolution

- For a relational database
 - Adding/removing a column, renaming an attribute
 - Known as database reorganization
- For a knowledge graph
 - Adding/removing a class
 - Adding/removing a superclass
 - Adding/removing a property
 - Adding/removing a constraint

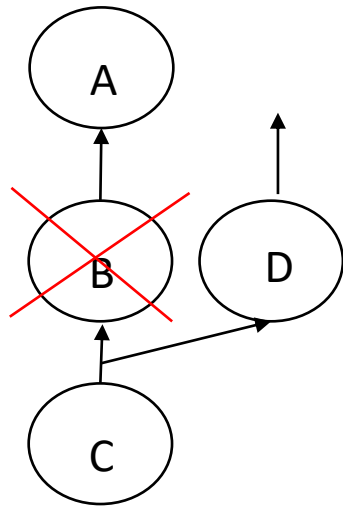
Approach is to maintain invariants, and make system-specific decisions

Schema Evolution

- Remove/rename a property
 - The change must be propagated through the graph
 - Generate a summary for review by the user

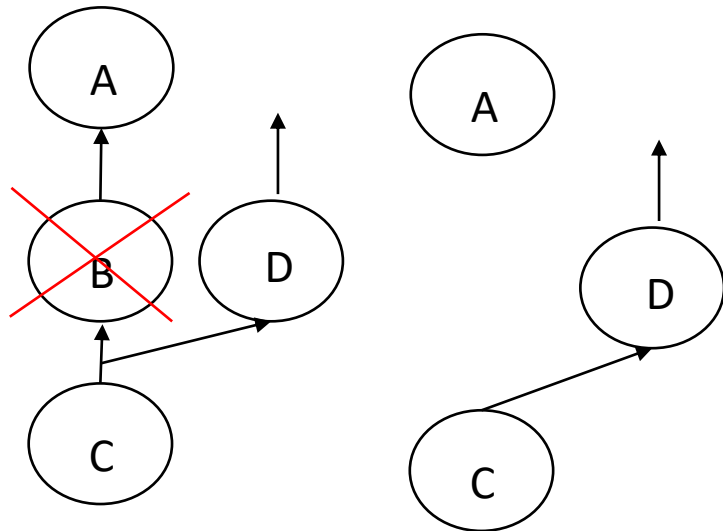
Schema Evolution

- Remove a class
 - What to do about its subclasses and instances
 - If its subclass has another parent, do nothing
 - Otherwise, make it a subclass of immediate parent
 - A more extreme step is to delete the subclasses and instances



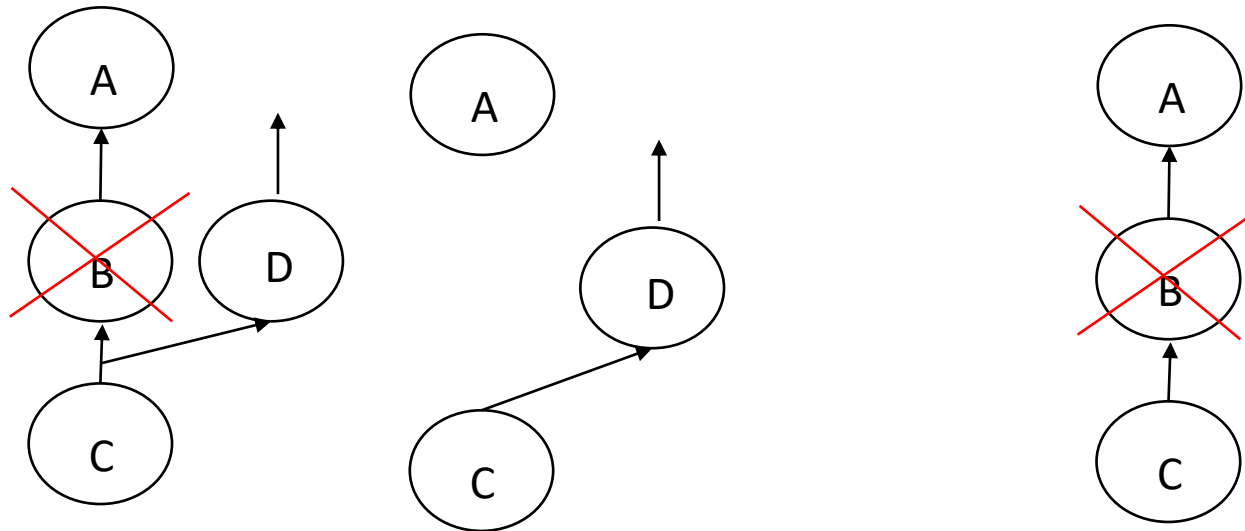
Schema Evolution

- Remove a class
 - What to do about its subclasses and instances
 - If its subclass has another parent, do nothing
 - Otherwise, make it a subclass of immediate parent
 - A more extreme step is to delete the subclasses and instances



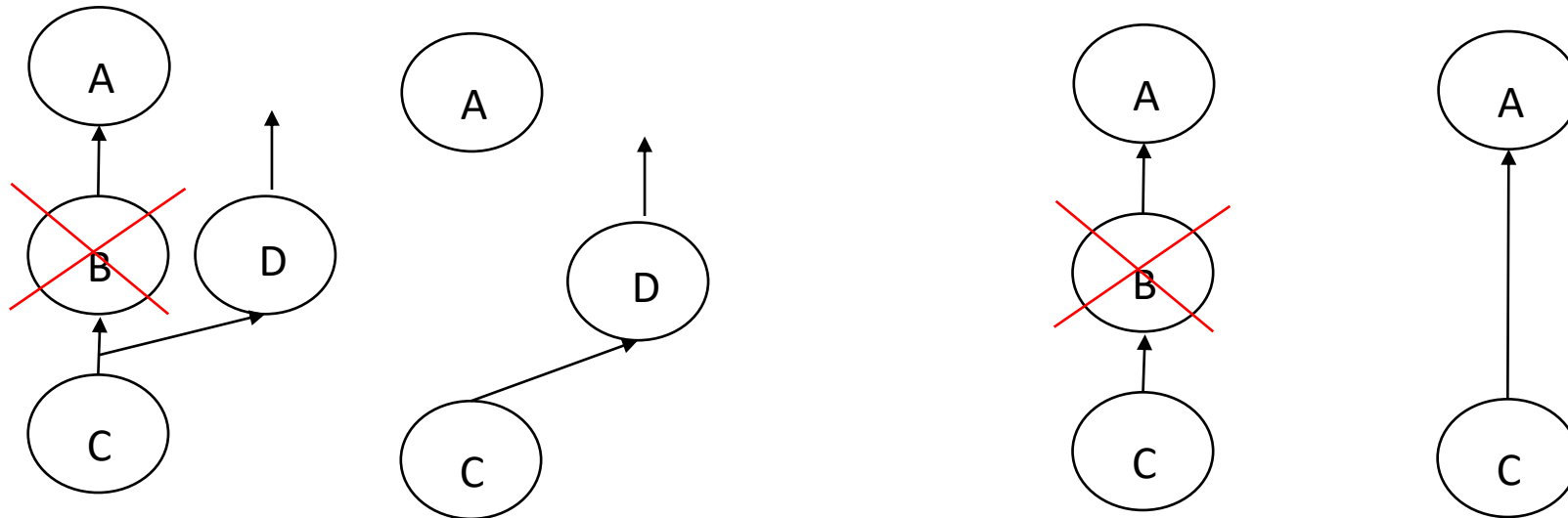
Schema Evolution

- Remove a class
 - What to do about its subclasses and instances
 - If its subclass has another parent, do nothing
 - Otherwise, make it a subclass of immediate parent
 - A more extreme step is to delete the subclasses and instances



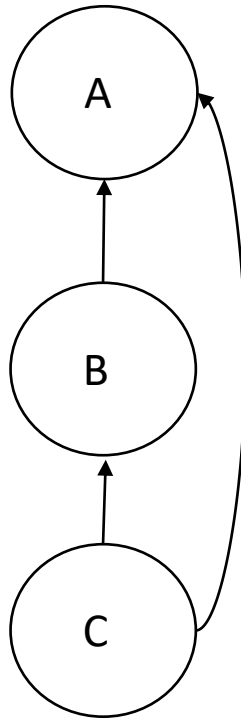
Schema Evolution

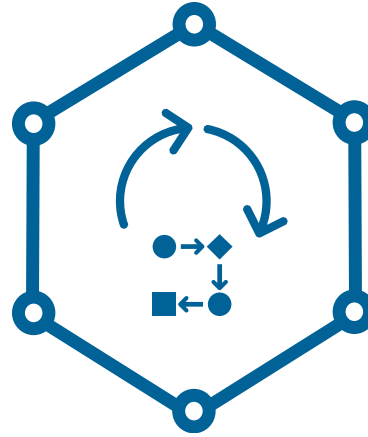
- Remove a class
 - What to do about its subclasses and instances
 - If its subclass has another parent, do nothing
 - Otherwise, make it a subclass of immediate parent
 - A more extreme step is to delete the subclasses and instances



Schema Evolution

- Add a super class
 - How to handle redundant links?
 - How to handle cycles?





Knowledge Graph Evolution Schema Evolution

Bonus

Emanuel Sallinger